

Preface

About OpenStax

OpenStax is part of Rice University, which is a 501(c)(3) nonprofit charitable corporation. As an educational initiative, it's our mission to improve educational access and learning for everyone. Through our partnerships with philanthropic organizations and our alliance with other educational resource companies, we're breaking down the most common barriers to learning. Because we believe that everyone should and can have access to knowledge.

About OpenStax Resources

Customization

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Instructors also have the option of creating a customized version of their OpenStax book. Visit the Instructor Resources section of your book page on OpenStax.org for more information.

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Errata

All OpenStax textbooks undergo a rigorous review process. However, like any professional-grade textbook, errors sometimes occur. Since our books are web-based, we can make updates periodically when deemed pedagogically necessary. If you have a correction to suggest, submit it through the link on your book page on OpenStax.org. Subject matter experts review all errata suggestions. OpenStax is committed to remaining transparent about all updates, so you will also find a list of past and pending errata changes on your book page on OpenStax.org.

Format

You can access this textbook for free in web view or PDF through OpenStax.org, and for a low cost in print.

About *Contemporary Mathematics*

Contemporary Mathematics is designed to meet the requirements for a liberal arts mathematics course. The textbook covers a range of topics that are typically found in a liberal arts course as well as some topics to connect mathematics to the world around us. The text provides stand-alone sections with a focus on showing relevance in the features as well as the examples, exercises, and exposition.

Pedagogical Foundation

Learning Objectives

Every section begins with a set of clear and concise learning objectives, which have been thoroughly revised to be both measurable and more closely aligned with current teaching practice. These objectives are designed to help the instructor decide what content to include or assign and to guide student expectations of learning. After completing the section and end-of-section exercises, students should be able to demonstrate mastery of the learning objectives.

Key Features

Check Your Understanding: Concept checks to confirm students understand content at the end of every section immediately before the exercise sets are provided to help bolster confidence before embarking on homework.

People in Mathematics: A mix of historic and contemporary profiles aimed to incorporate extensive diversity in gender and ethnicity. The profiles incorporate how the person's contribution has benefitted students or is relevant to their lives in some way.

Who Knew?: A high-interest feature designed to showcase something interesting related to the section contents. These features are crafted to offer something students might be surprised to find is so relevant to them.

Work It Out: Offers some activity ideas in line with the sections to support the learning objectives.

Tech Check: Highlights technologies that support content in the section.

Projects: A feature designed to put students in the driver's seat researching a topic using various online resources. It is intended to be primarily or wholly non-computational. Projects utilize online research and writing to summarize their findings.

Section Summaries

Section summaries distill the information in each section for both students and instructors down to key, concise points addressed in the section.

Key Terms

Key terms are bold and are followed by a definition in context.

Answers and Solutions to Questions in the Book

Answers for Your Turn and Check Your Understanding exercises are provided in the Answer Key at the end of the book. The Section Exercises, Chapter Reviews, and Chapter Tests are intended for homework assignments or assessment; thus, student-facing solutions are provided in the Student Solution Manual for only a subset of the exercises. Solutions for all exercises are provided in the Instructor Solution Manual for instructors to share with students at their discretion, as is standard for such resources.

About the Authors

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Donna Kirk received her B.S. in Mathematics from the State University of New York at Oneonta and her master's degree from City University – Seattle in Educational Technology and Curriculum Design. After teaching math in higher education for more than twenty years, she joined University of Wisconsin's Education Department in 2021, teaching math education for teacher preparation. She is also the director of a STEM institute focused on connecting underrepresented students with access to engaging and innovative experiences to empower themselves to pursue STEM related careers.

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Additional Resources

Student and Instructor Resources

We've compiled additional resources for both students and instructors, including student solution manuals, instructor solution manuals, and PowerPoint lecture slides. Instructor resources require a verified instructor account, which you can apply for when you log in or create your account on OpenStax.org. Take advantage of these resources to supplement your OpenStax book.

Academic Integrity

Academic integrity builds trust, understanding, equity, and genuine learning. While students may encounter significant challenges in their courses and their lives, doing their own work and maintaining a high degree of authenticity will result in meaningful outcomes that will extend far beyond their college career. Faculty, administrators, resource providers, and students should work together to maintain a fair and positive experience.

We realize that students benefit when academic integrity ground rules are established early in the course. To that end, OpenStax has created an interactive to aid with academic integrity discussions in your course.



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Visit our [academic integrity slider \(https://openstax.org/r/academic-integrity-slider\)](https://openstax.org/r/academic-integrity-slider). Click and drag icons along the continuum to align these practices with your institution and course policies. You may then include the graphic on your syllabus, present it in your first course meeting, or create a handout for students.

At OpenStax we are also developing resources supporting authentic learning experiences and assessment. Please visit this book's page for updates. For an in-depth review of academic integrity strategies, we highly recommend visiting the [International Center of Academic Integrity \(ICAI\) website \(https://openstax.org/r/academicintegrity\)](https://openstax.org/r/academicintegrity).

Community Hubs

OpenStax partners with the Institute for the Study of Knowledge Management in Education (ISKME) to offer Community Hubs on OER Commons—a platform for instructors to share community-created resources that support OpenStax books, free of charge. Through our Community Hubs, instructors can upload their own materials or download resources to use in their own courses, including additional ancillaries, teaching material, multimedia, and relevant course content. We encourage instructors to join the hubs for the subjects most relevant to your teaching and research as an

opportunity both to enrich your courses and to engage with other faculty. To reach the Community Hubs, visit www.oercommons.org/hubs/openstax (<https://openstax.org/r/community-hub>).

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